

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **6700**

L 61, 482, 986, 1, 1505, 48245

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	202	Seat No:	2022600004
		Course Section:	1		



Name of the candidate. Bande Aniket Pandit

Candidate's UID number:

2	0	2	2	6	0	0	0	4
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Examination class room number:

2	0	2
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Invigilator's signature with date:

	08/10/2025
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(To be filled in by the candidate)

EXAMINATION: BE B.Tech MSE
(For example FY MCA/FY BTECH)

SEMESTER: VII
✓

PROGRAM: CSE - Aimd

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08/10/25 Wednesday

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks

Question No.

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(Begin answer for each question on a new page)

Q.1.

b) $\Rightarrow$ overfitting is when the model learns the data to that extent that it starts learning the errors also. So the model becomes too complex for the given data. Hence when model has ~~has~~ high capacity the overfitting occurs. It happens when there is high variance in data. How it ~~effe~~ affects the generalization? Due to overfitting model gets error prone. With simple data too it sometimes give incorrect result.

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Q.2.

b) $\Rightarrow$ A autoencoder is a artificial neural network trained with efficient data. It is mainly used to learn the features from data or to do the dimensionality reduction.

Linear undercomplete autoencoder is the encoder who has ~~number~~ ~~of~~ ~~a~~ narrow bottleneck than the input data. Basically here the network forces the ~~data~~ data to pass through narrow bottleneck.

The ~~auto~~encoder has to learn to ~~grab~~ grab important and silent features in this. Then the decoder will efficiently reconstruct the data on its own ability.

For the numeric data, we generally use the MSE as a loss function which is difference between the reconstructed output & the input data. It is also called a reconstruction error. The main aim of linear undercomplete autoencoder is to minimize this error.

If we have the bottleneck very much narrower. Then encoder may loose the important information due to which decoder will fail to generate the output correctly & we will have high reconstruction error.

This linear undercomplete autoencoder is a classical ~~and~~ ~~artifice~~ ^{or} artifice for the dimensionality ~~&~~ reduction as with effective bottleneck narrowness we can reduce the dimensionality of input data just like we do in PCA (Principal component analysis).

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In this way the linear under complete autoencoders trained with MSE learns a subspace equivalent to PCA for dimensionality reduction of given data.

over complete autoencoder without regularization is the one in which the bottleneck or latent space has same or more number of dimensions than the input. Which means encoder ~~is~~ just copies the input & does not learn any features. It just tends to learn the identity function in this way.

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Q.3.

a) $\Rightarrow$

RNN (Recurrent neural network) is used for the data where we want the sequence to be considered. It is very good with easy, simple & small data. But when it comes to the large ~~data~~ length data it just forgets the context of earlier used data. How?

In RNN the activation ~~of~~ of ~~current~~ current state is trained with the previous one but when there is a longer data the activation or context becomes very low or ≈ 0 so it just forgets about the earlier data context.

$$\frac{\partial L}{\partial w_i} = \sum \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_i} \times \frac{\partial a_i}{\partial w_i}$$

$$o = h \cdot t - k$$

$$\text{i.e. } \frac{\partial L}{\partial w_i} = \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_1} \times \frac{\partial a_1}{\partial w} + \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_2} \times \frac{\partial a_2}{\partial w}$$

$$\text{-----} \frac{\partial L}{\partial o} \times \frac{\partial o}{\partial a_{100}} \times \frac{\partial a_{100}}{\partial w}$$

This uses sigmoid function whose value ranges in 0 to 1.

multiplication of many such values make the result $\frac{\partial L}{\partial w} \approx 0$.

i.e. it loses context with larger duration.

This is known as the ~~gradient~~ ~~des~~ gradients vanishing problem


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The LSTM solves the gradient vanishing problem. It ~~has~~ has long memory context due to the memory cell state along with ~~the~~ the hidden state. How it does that?

LSTM has 3 gates

- Forget gate
- Input gate
- Output gate

The forget gate decides which info from the previous state to forget. The input gate decides which input new info to take & output gate decide what to output.

The use of forget gate to retain important info plays very crucial role to make the LSTM to remember contexts for a longer data also.

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Q.3.

b) $\Rightarrow$ GRU is gated recurrent unit, it is a simpler version of LSTM. It also solves the problem of gradient vanishing like LSTM does -

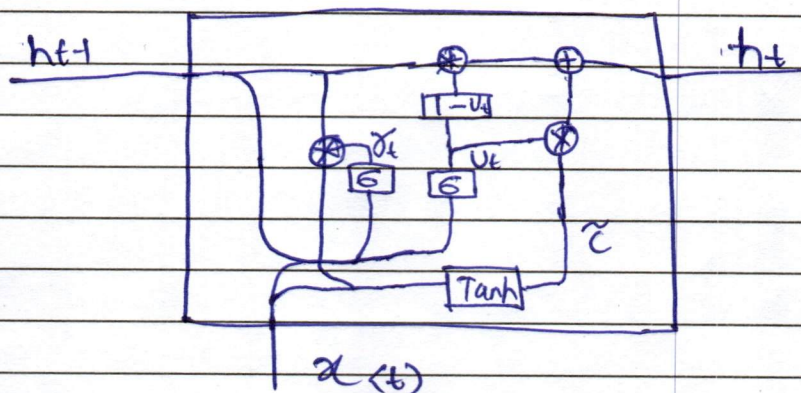
In GRU the hidden and memory cell state is combined. ~~to~~ which makes it little simpler. It has 2 gates.

$\rightarrow$ update gate

$\rightarrow$ reset gate.

Update gate decides which info from previous to remember & which info from current state to take into consideration. While the reset gate to forget the thing which are ~~states~~ unnecessary.

Following are GRU architecture & equations.



$$h_t = u_t * \tilde{c} + (1 - u_t) h_{t-1}$$

$$\tilde{c} = \tanh(W_c (r_t * h_{t-1}, X) + b_c)$$

$$u_t = \sigma(W_u (h_{t-1}, X) + b_u)$$

$$r_t = \sigma(W_r (h_{t-1}, X) + b_r)$$



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Question
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As GRU has less number of states & Gates than the LSTM is its little simpler, less complex than the LSTM. But it is comparability quite similar to LSTM to remember longer context. Its due to the update & reset gate. Its due to the combined state of memory cell state & candidate hidden state.

When we need efficiency and faster results we prefer to go with GRU. As it requires less computations due to less complex structure.

LSTM is better when we need to be very, very accurate & need more control on inputs & learnings. LSTM is for medium & small datasets with greater control. While GRU is fast & efficient for the larger datasets.

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Q.2.

a) $\Rightarrow$

Computational complexity (number of multiplications).

for standard 2D convolution.
its.

$$H_{out} = \left\lfloor \frac{H_{in} - F + 2P}{s} \right\rfloor + 1$$

Floor value.

where the P is padding applied & s are stride.

Similarly.

$$W_{out} = \left\lfloor \frac{W_{in} - F + 2P}{s} \right\rfloor + 1$$

$\therefore$ The output size for std 2D conv is. Here.

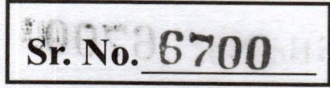
$$(H - F + 1) \times (W - F + 1) \times K.$$

as there are K filters.

for depthwise separable convolutions.
its.

$$(H - F + 1) \times (W - F + 1) \times K \times C.$$

[illegible]



Sr. No. 6700

[illegible]

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